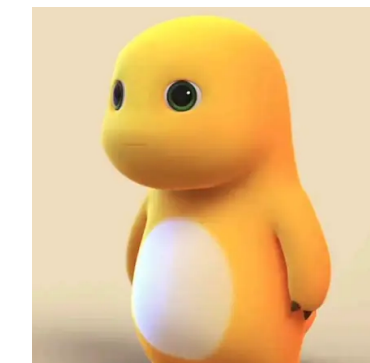
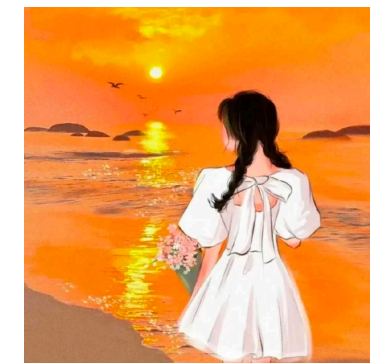


LaMET-Agent: Agentic Workflow for Large-Momentum Effective Theory Analysis

2026.07.20

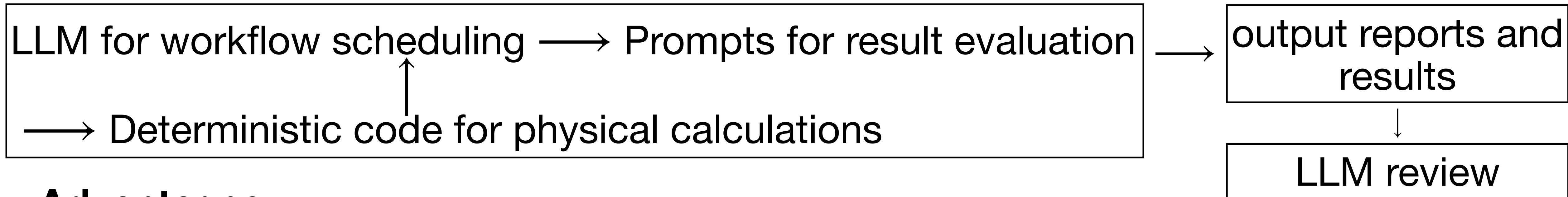
GPT, DeepSeek, Jinchen He, Xiangyu Jiang, Fei Yao and Dian-Jun Zhao



LLM: find certainty in uncertainty

<https://github.com/Greyyy-HJC/lamet-agent/tree/main>

What can LLM do in a (relatively defined) numerical analysis workflow?



Advantages:

1. Frees users from tedious numerical evaluation processes;
2. Avoids subjective assumptions in deterministic numerical analysis using LLM;
3. Saves LLM tokens.

Disadvantages:

I am thinking about how to widen more computational limitations (using moment as an example)

Cannot consider stories outside of LaMET framework.

LaMET-Agent: project structure

<https://github.com/Greyyy-HJC/lamet-agent/tree/main>

1. **CLI reads .json/.jsonc manifest:** `src/lamet_agent/cli.py`
2. **API schedules stages:** `src/lamet_agent/agent.py`
3. **Prompt decides to call LLM + tool:** `src/lamet_agent/stages/*/prompts.py`

For the LaMET, stages mainly consist of: **correlator_analysis, renormalization, fourier_transform, perturbative_matching, extrapolation, review.**

4. **Tool determines physical calculation and output report:** `src/lamet_agent/stages/*/{functions.py, report.py}`

Users can complete entire LaMET workflow using only correlator + json scripts.
`examples/{sample_manifest.jsonc, partial_sample_manifest.jsonc}`

LaMET-Agent: current status

Stages	Completed	Incomplete	By
data structure	h5 file to input the correlator, .nc file to output for all stages, jackknife, bootstrap or also bin-sample	/	JXY
plan	Convert npy/npz/h5 to the input correlator h5 file through dialogue with LLM, check if necessary parameters exist, etc	/	HJC/ ZDJ
correlator_analysis (Parallel processing for sample fits)	2/3-state joint fitting for forward/non-forward, CG/GI, PP/PS/SS BME with BMA using DARatio/Ratio/FH	GEVP, Lanzos, non-forward FH	HJC/ ZDJ/JXY
renormalization	Ratio/Hybrid-Ratio/Self	/	HJC
fourier_transform	Asymptotic extrapolation based on arXiv: 2601.12189 with BMA, CG/GI hadron q/g PDF/DA/GPD	/	ZDJ
perturbative_matching	CG/GI gt/gtg5/gz/gzg5/gtgpg5 qPDF/qDA MSbar/Ratio/Hybrid NLO matching kernel	qGPD, qTMD, gluon operator, RGR	YF
extrapolation	/	To be continued...	All
review	LLM can summarize, diagnose, and adjust manifest based on reports and .nc file provided by each stage	/	ZDJ

LaMET-Agent: image output

Stages	Image output (.pdf/.svg)	
correlator_analysis	3pt_ratio: dispersion_relation, ca_p<pz>, ca_<ensemble>_{re/im}, fit_logs	qda_ratio: ca_p<pz>, ca_<ensemble>_{re/im}
renormalization	Hybrid-ratio: rn_p<pz>, rn_<ensemble>_{re/im}	Self: lnM_vs_inv_a, M_over_zR, mR_zmsbar,
fourier_transform	ft_p<pz>_{re/im}, ft_p<pz>_xdep, ft_<ensemble>_xdep	
perturbative_matching	mt_p<pz>, mt_<ensemble>	
extrapolation	To be continued...	

LaMET-Agent: perturbative_matching

```
"kernels": [  
  {  
    "stage": "perturbative_matching",  
    "kernel_id": "YOUR_OWN_KERNEL",  
    "kernel_path": "src/lamet_agent/kernels.py",  
    "scheme": "hybrid_ratio",  
    "kernel_parameters": {"zs_fm": 0.1722}  
  }  
]
```

**Distribution = GPD, Current = Gluon
is still in progress.**

There are many perturbation results, so users can
add kernel themselves: src/lamet_agent/kernels.py

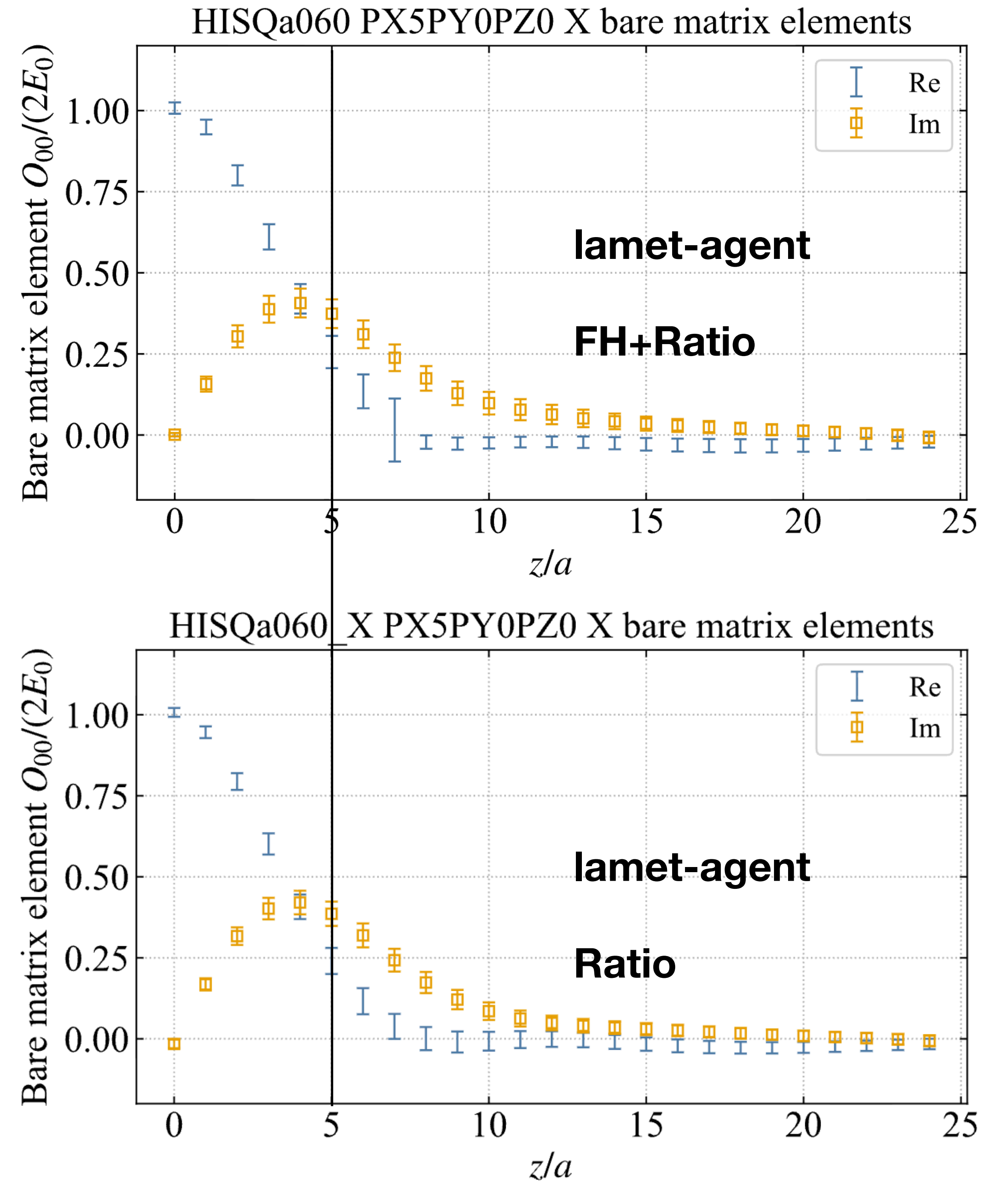
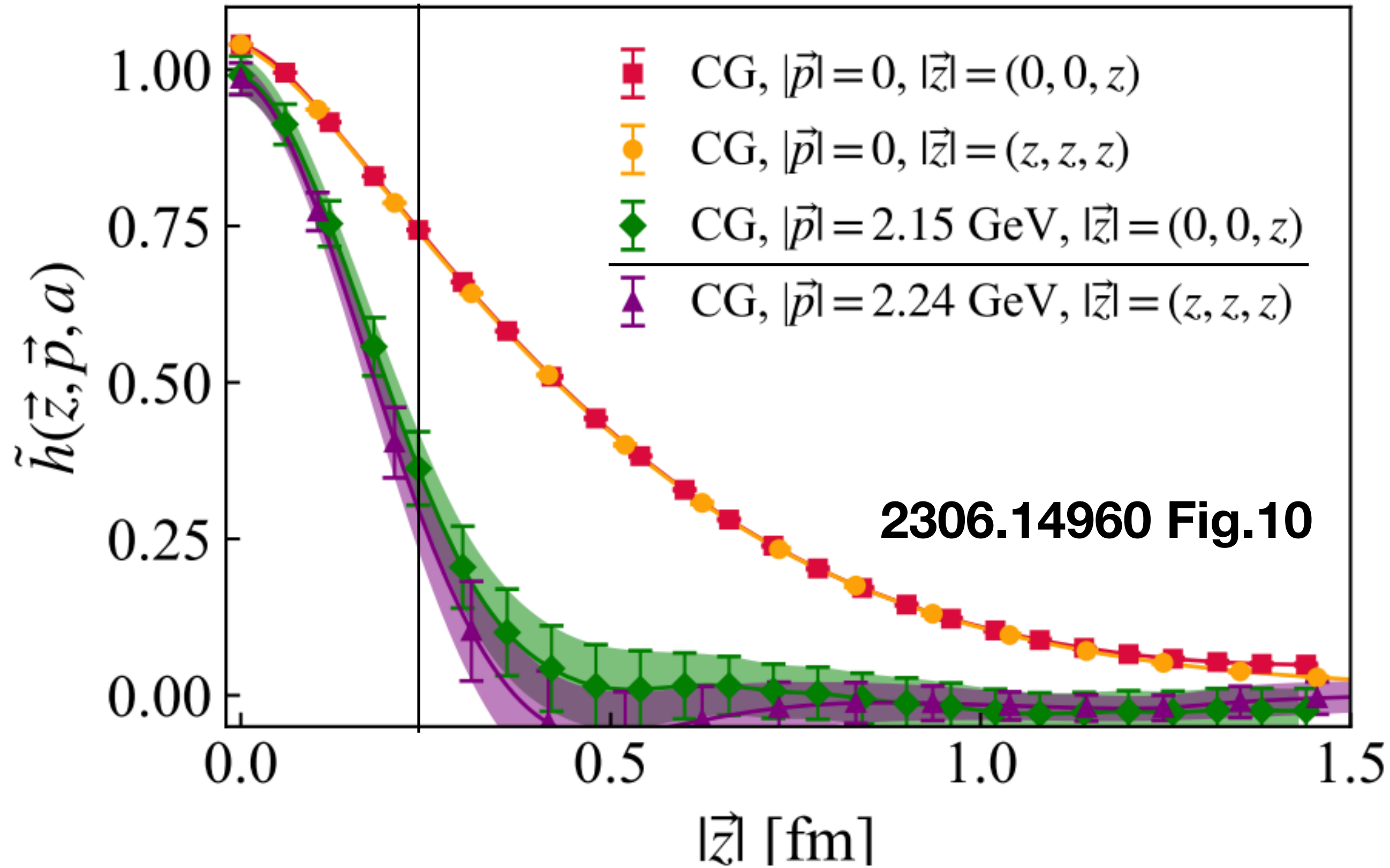
Supported matching kernels	Gauge	Operators	Distribution	Scheme	Order
arXiv: 2602.11283 2412.20461 2604.00143 2208.08008 2212.14415	Coulomb-Gauge	γ_t	quasi-PDF	MSbar	NLO
		$\gamma_t \gamma_5$		Ratio	
	Gauge-Invariant	γ_z	quasi-DA		
		$\gamma_z \gamma_5$		Hybrid-Ratio	
		$\gamma_t \gamma_\perp \gamma_5$			

LaMET-Agent: numerical test

<https://github.com/Greyyy-HJC/lamet-agent/tree/main>

CG/GI pion PDF, X. Gao, et al., *Phys.Rev.D* 109 (2024) 9, 094506

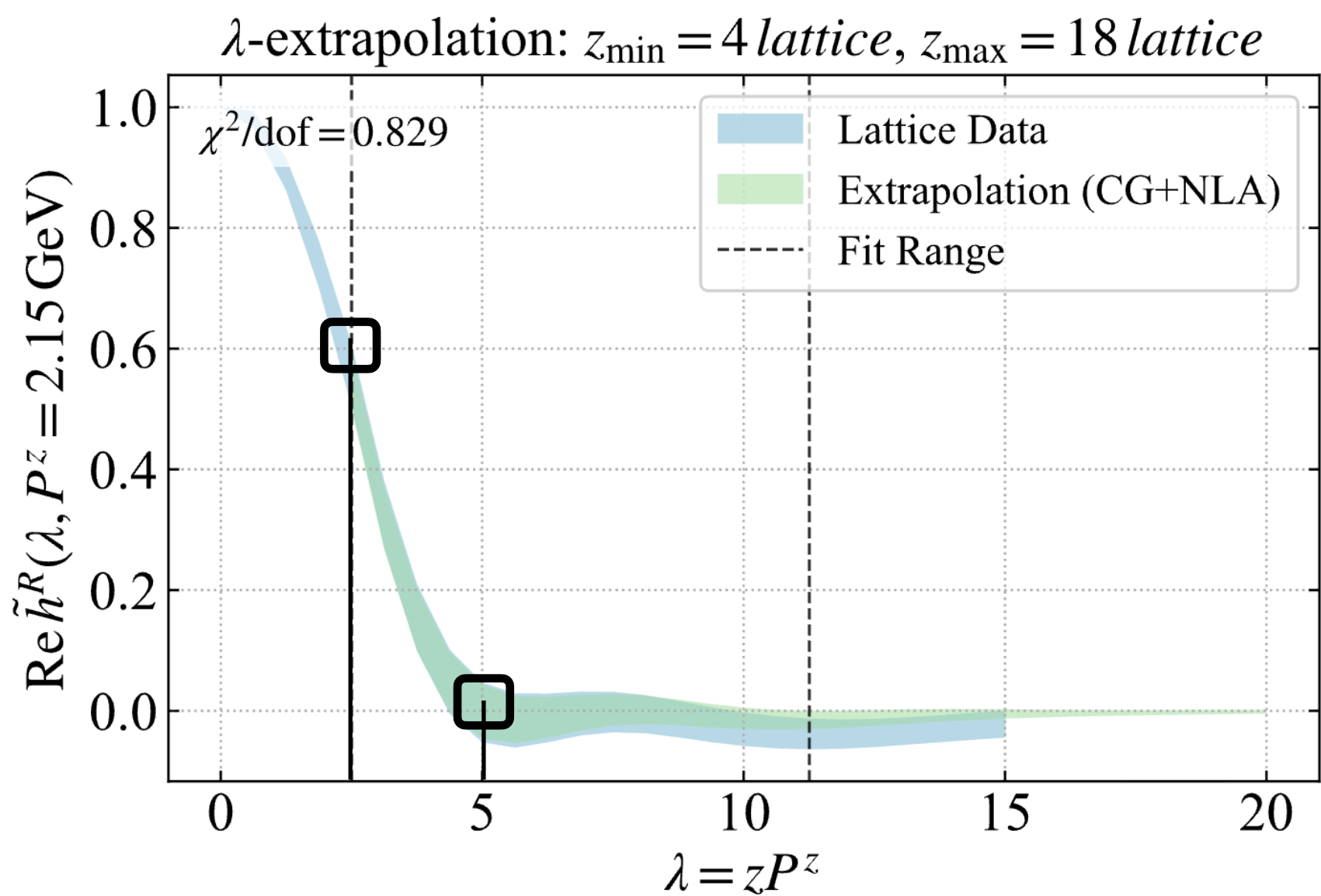
Correlator Analysis



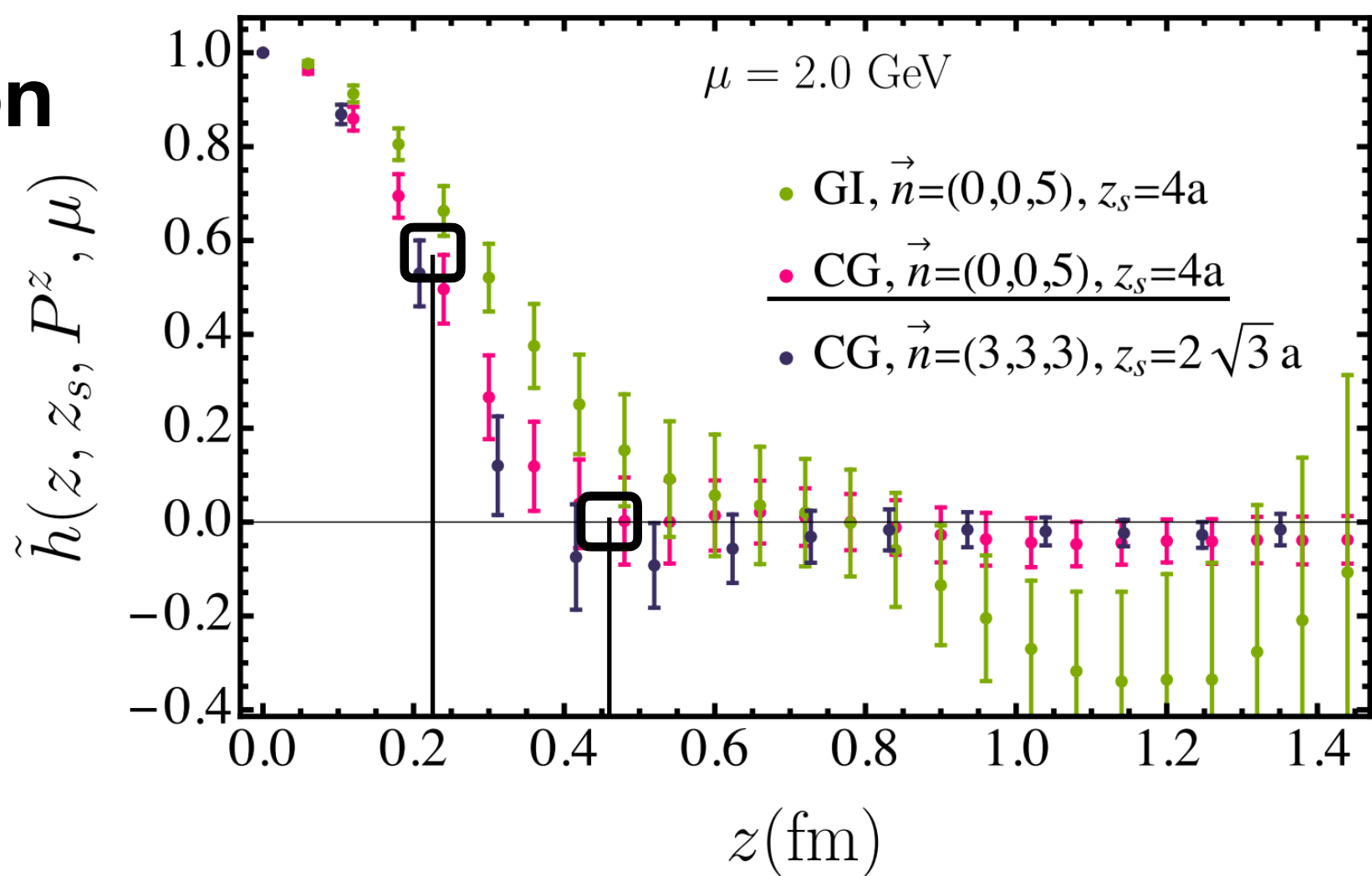
LaMET-Agent: numerical test

<https://github.com/Greyyy-HJC/lamet-agent/tree/main>

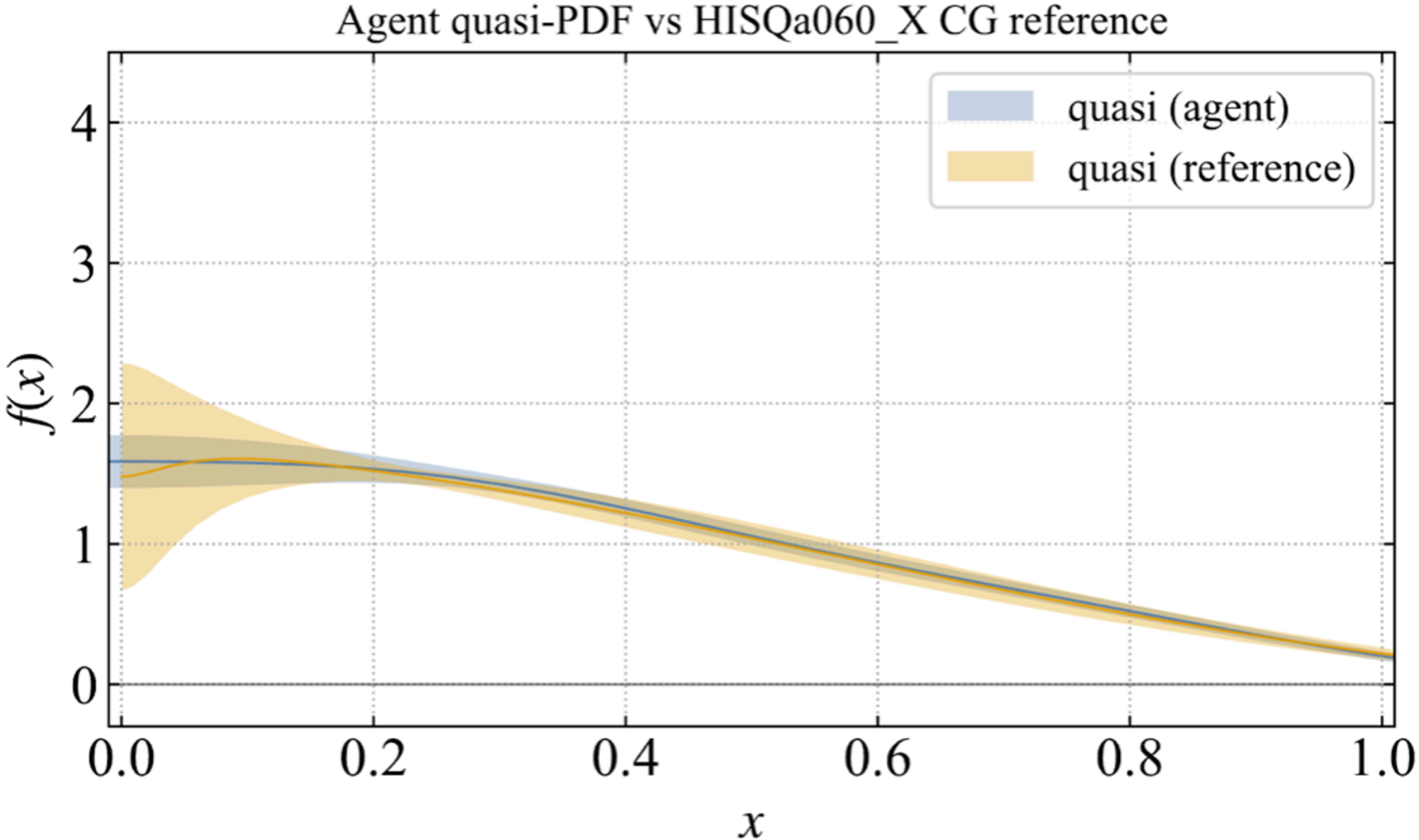
CG/GI pion PDF, X. Gao, et al., *Phys.Rev.D* 109 (2024) 9, 094506



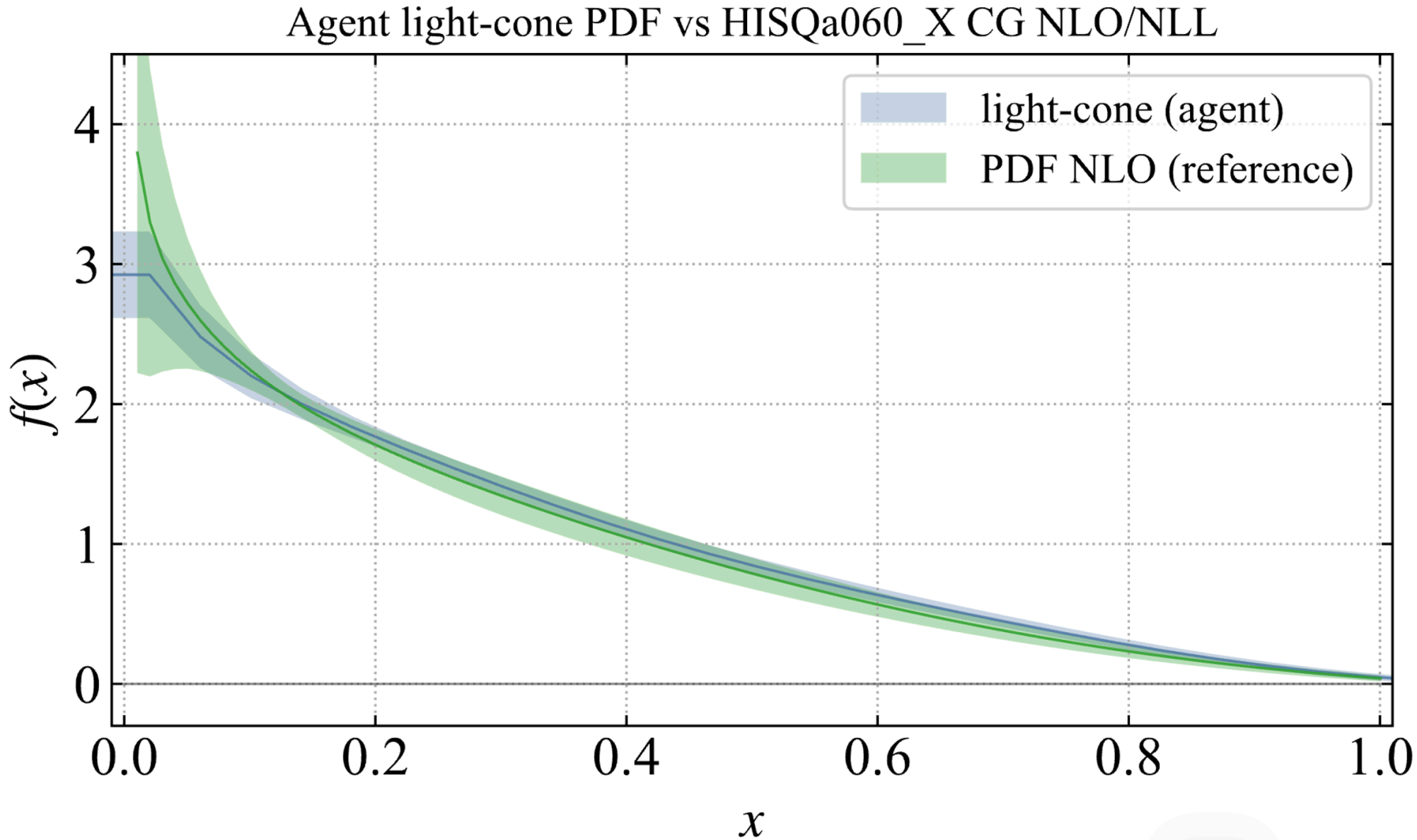
Hybrid Ratio Renormalization



Fourier Transform



Perturbative Matching

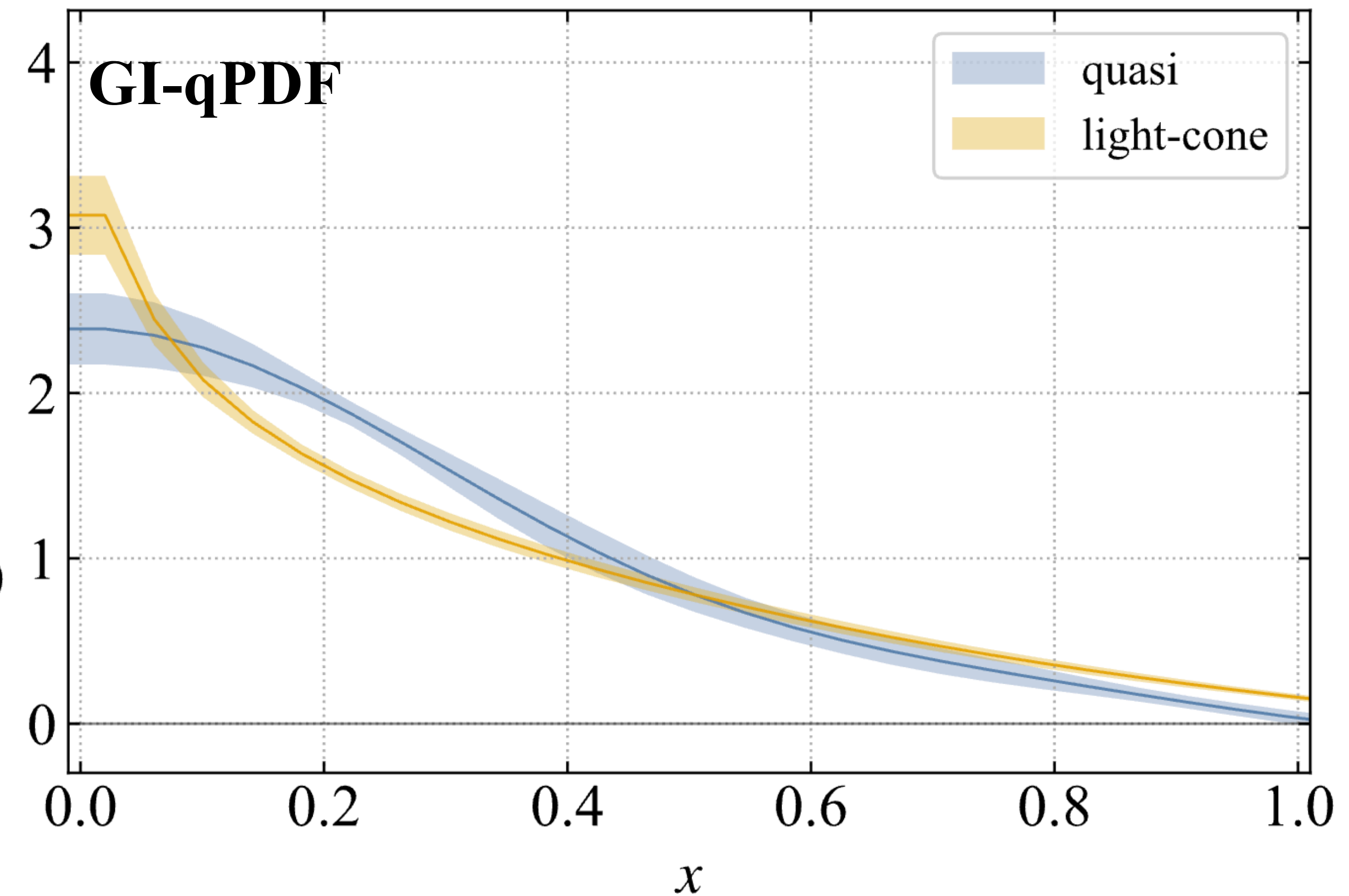
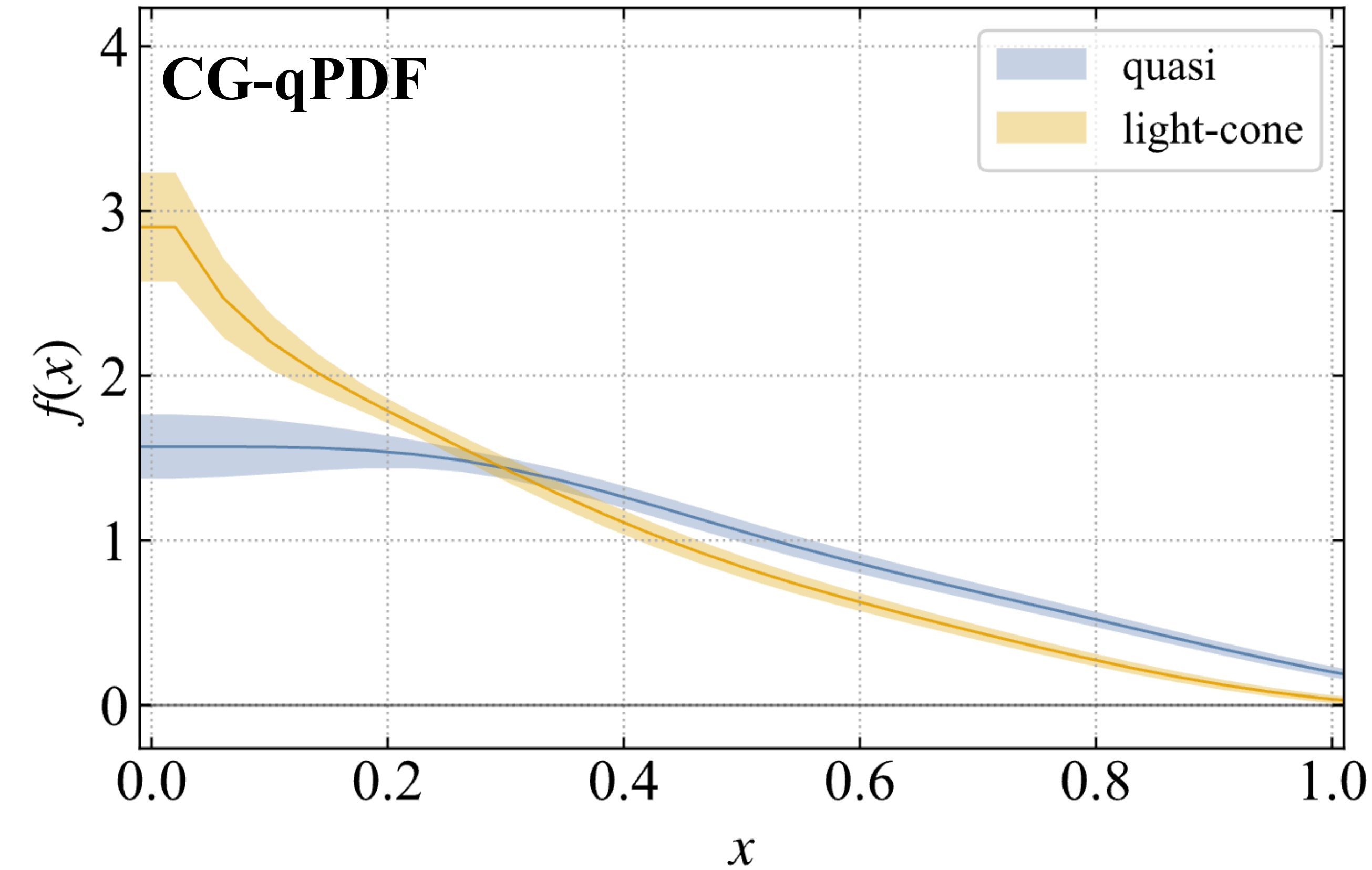


We are currently considering how to incorporate systematic errors into the entire stages.

LaMET-Agent: numerical test

<https://github.com/Greyyy-HJC/lamet-agent/tree/main>

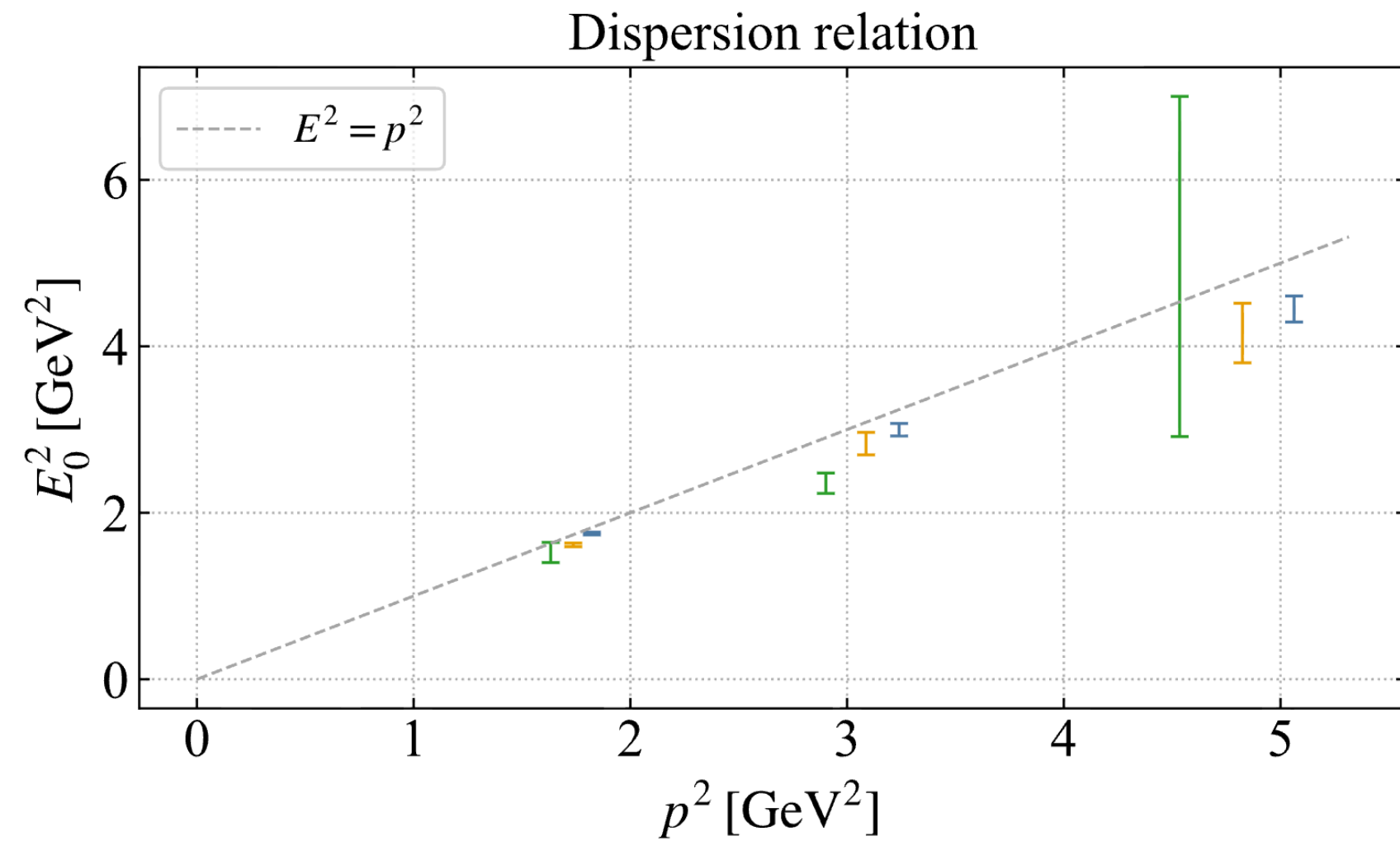
CG/GI pion PDF, X. Gao, et al., *Phys.Rev.D* 109 (2024) 9, 094506



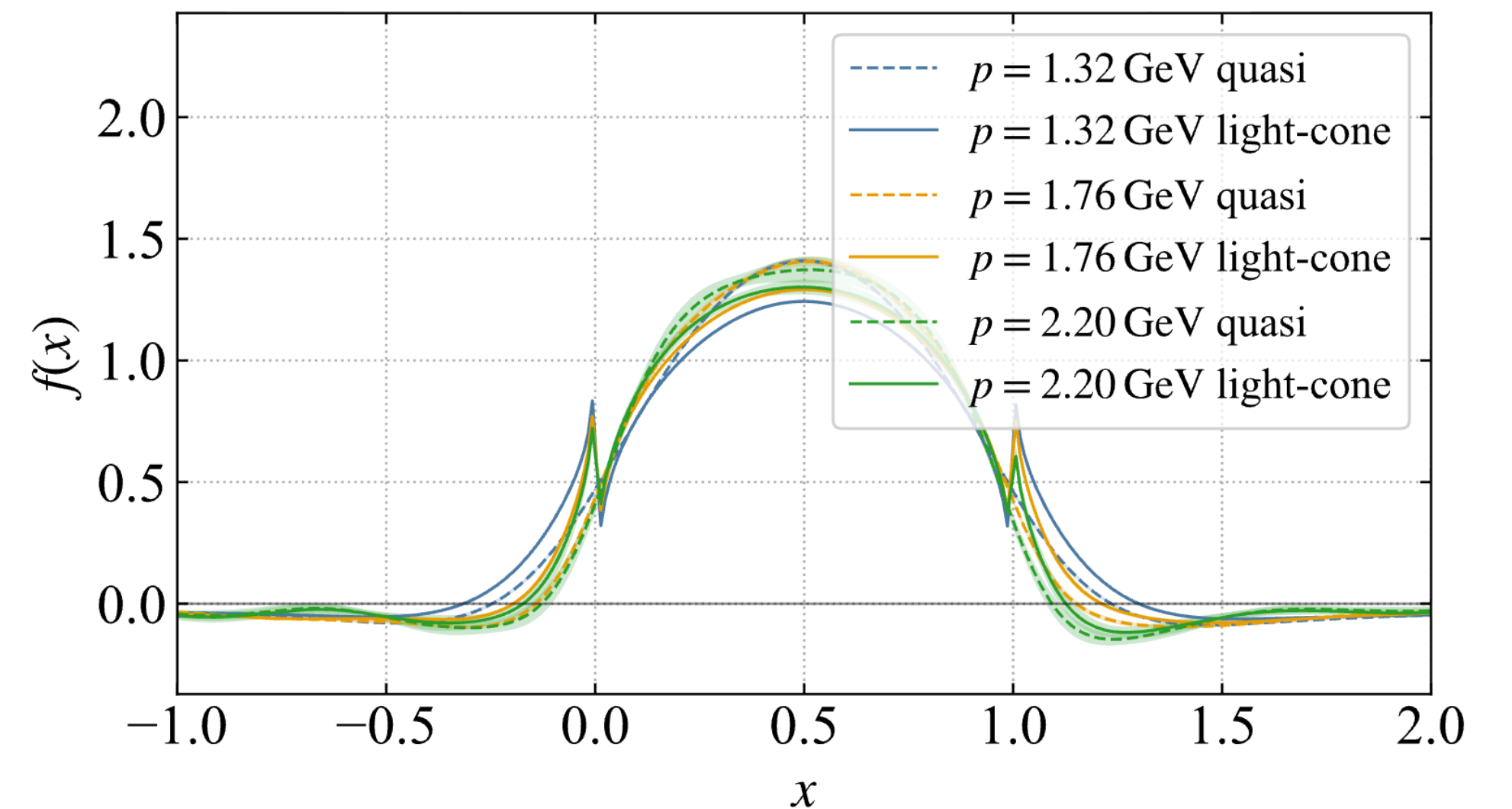
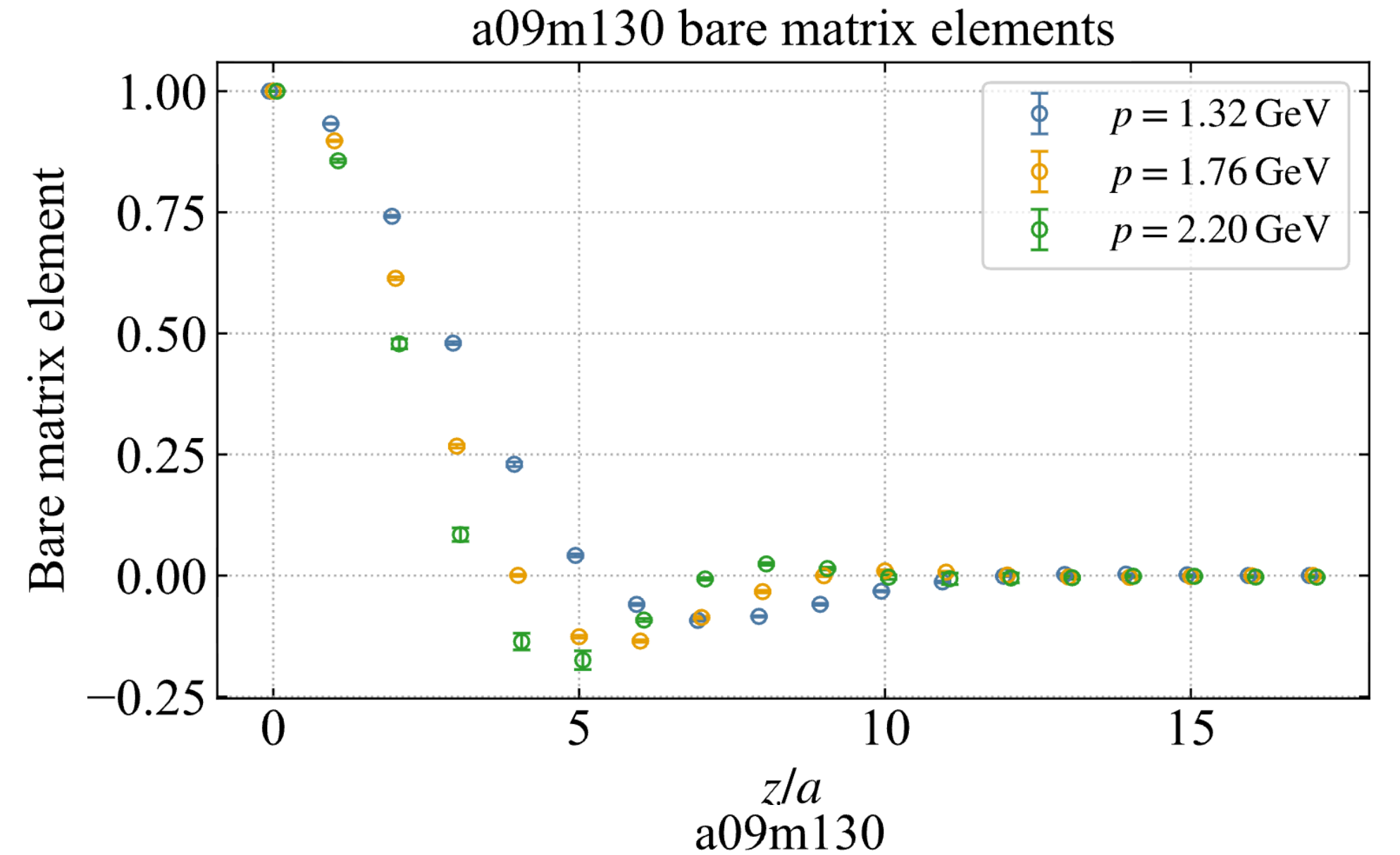
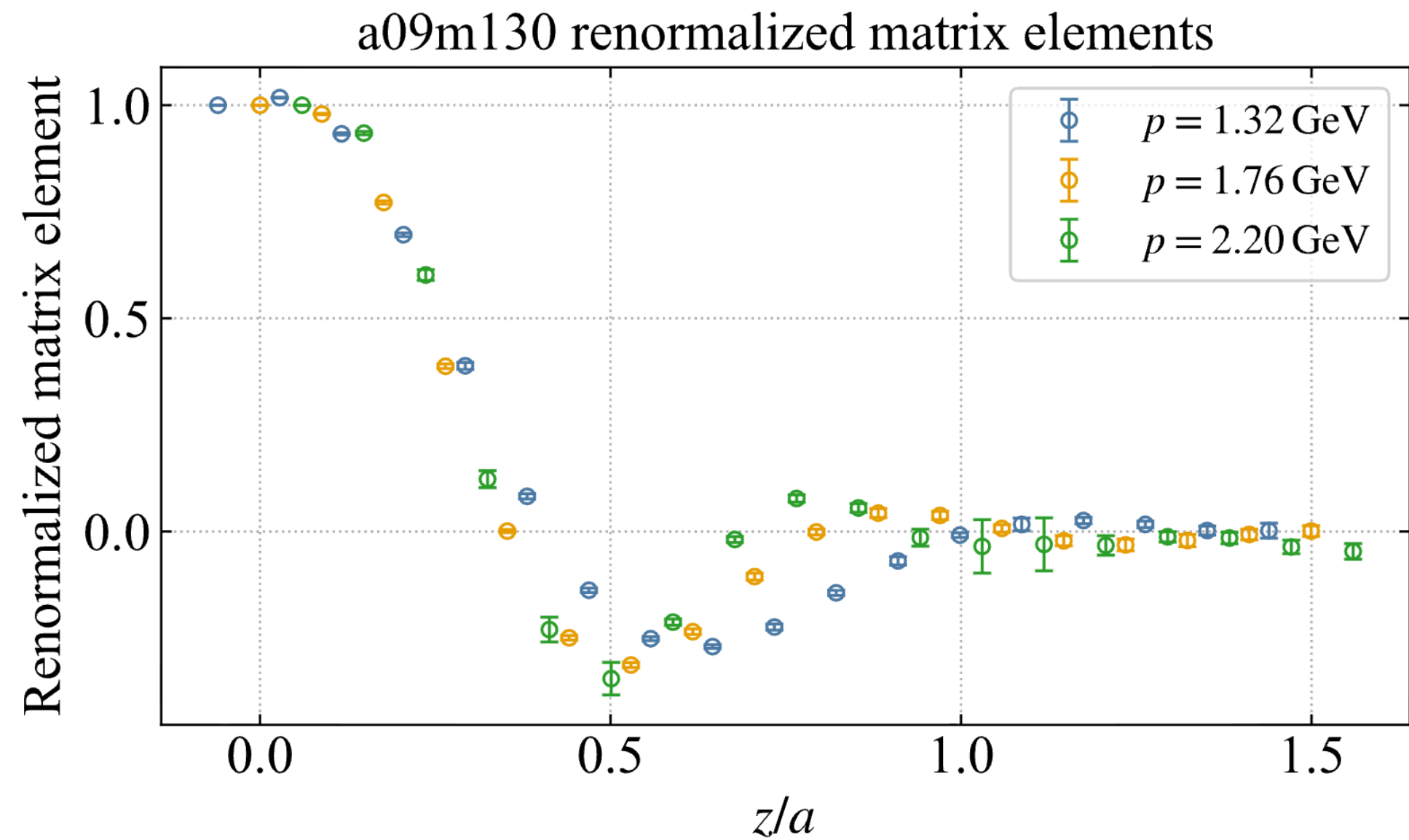
LaMET-Agent: numerical test

<https://github.com/Greyyy-HJC/lamet-agent/tree/main>

GI π/K DA, J. Hua, et al., *Phys.Rev.Lett.* 129 (2022) 13, 132001



Pion case



LaMET-Agent: review

```
"stages": ["correlator_analysis", "renormalization", "fourier_transform", "perturbative_matching", "review"]

"review": {
  "defaults": {},
  "jobs": [{"id": "review"}]
}
```

Review 的输入材料来自：<root_directory>/<artifacts_directory>/<stage>/，只要 stage 子目录存在，或 manifest metadata.stages 声明了该 stage，就会纳入材料包。

每个 **stage** 收集： 1. stage report; 2. NetCDF 文件; 3. SVG 路径：

输出：<artifacts_directory>/review/review.md

review prompt 要求每个 **stage** 写：Physical Summary, Key figure, Diagnostics, Recommended Manifest Changes

LaMET 格点

1. Correlator analysis

Physical Summary

Key figure

Diagnostics

Recommended Manifest Changes

2. Renormalization

Physical Summary

Key figure

Diagnostics

Recommended Manifest Changes

3. Fourier transform

Physical Summary

Key figure

Diagnostics

Recommended Manifest Changes

4. Perturbative matching

Physical Summary

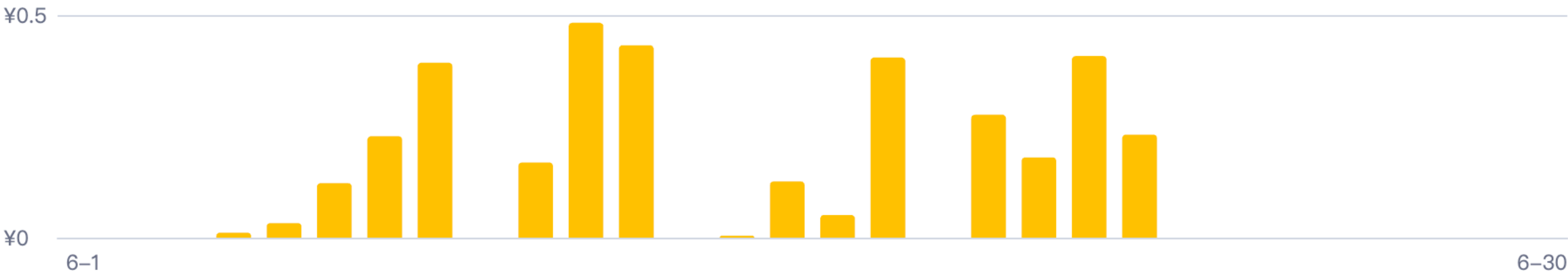
中文：
关联函数分析。 我们从 HISQ 系综 ($a = 0.0574$ fm) 上的 π 介子两点和三点关联函数中提取了裸矩阵元 $h(z, P_z)$ 。对零动量 ($P_z = 0$) 和 $P_z = 2.15$ GeV 两个通道分别进行了 Breit ratio 联合拟合，采用 $N_{\text{st}} = 2$ 的谱分解模型，拟合窗口为 $t \in [4, 12)$ (零动量) 和 $t \in [2, 12)$ ($P_z = 2.15$ GeV), $t_{\text{sep}} = 8, 10, 12$, $\tau_{\text{cut}} = 2$ 。所有 z 点的拟合质量良好：零动量通道的 χ^2/dof 范围为 0.75–1.07, $P_z = 2.15$ GeV 通道为 0.55–0.87, 且所有 109 个 jackknife 样本均未出现拟合失败。零动量通道的 Q 值从 $z = 0$ 的 0.88 单调下降至 $z = 19$ 的 0.54, 表明大 z 处信噪比有所降低但仍可接受; $P_z = 2.15$ GeV 通道的 Q 值整体更高 (0.73–1.00), 显示出更好的拟合稳定性。裸矩阵元 $h(z)$ 的实部在 $z = 0$ 处归一化至约 1.0, 随 z 增大呈指数衰减, 虚部在统计误差内接近零, 符合预期行为。

English:
Correlator analysis. We extracted bare matrix elements $h(z, P_z)$ from pion two-point and three-point correlation functions on the HISQ ensemble ($a = 0.0574$ fm). Breit ratio joint fits were performed for both the zero-momentum ($P_z = 0$) and $P_z = 2.15$ GeV channels, employing a spectral decomposition model with $N_{\text{st}} = 2$, fit windows $t \in [4, 12)$ (zero momentum) and $t \in [2, 12)$ ($P_z = 2.15$ GeV), $t_{\text{sep}} = 8, 10, 12$, and $\tau_{\text{cut}} = 2$. The fit quality is satisfactory across all z values: χ^2/dof ranges from 0.75 to 1.07 for the zero-momentum channel and from 0.55 to 0.87 for the $P_z = 2.15$ GeV channel, with zero fit failures among all 109 jackknife samples. The Q -value for the zero-momentum channel decreases monotonically from 0.88 at $z = 0$ to 0.54 at $z = 19$, indicating a moderate signal-to-noise degradation at large z that remains acceptable; the $P_z = 2.15$ GeV channel exhibits higher Q -values (0.73–1.00), demonstrating superior fit stability. The real part of the bare matrix element $h(z)$ is normalized to approximately 1.0 at $z = 0$ and decays exponentially with z , while the imaginary part is consistent with zero within statistical uncertainties, as expected.

LaMET-Agent: API usage

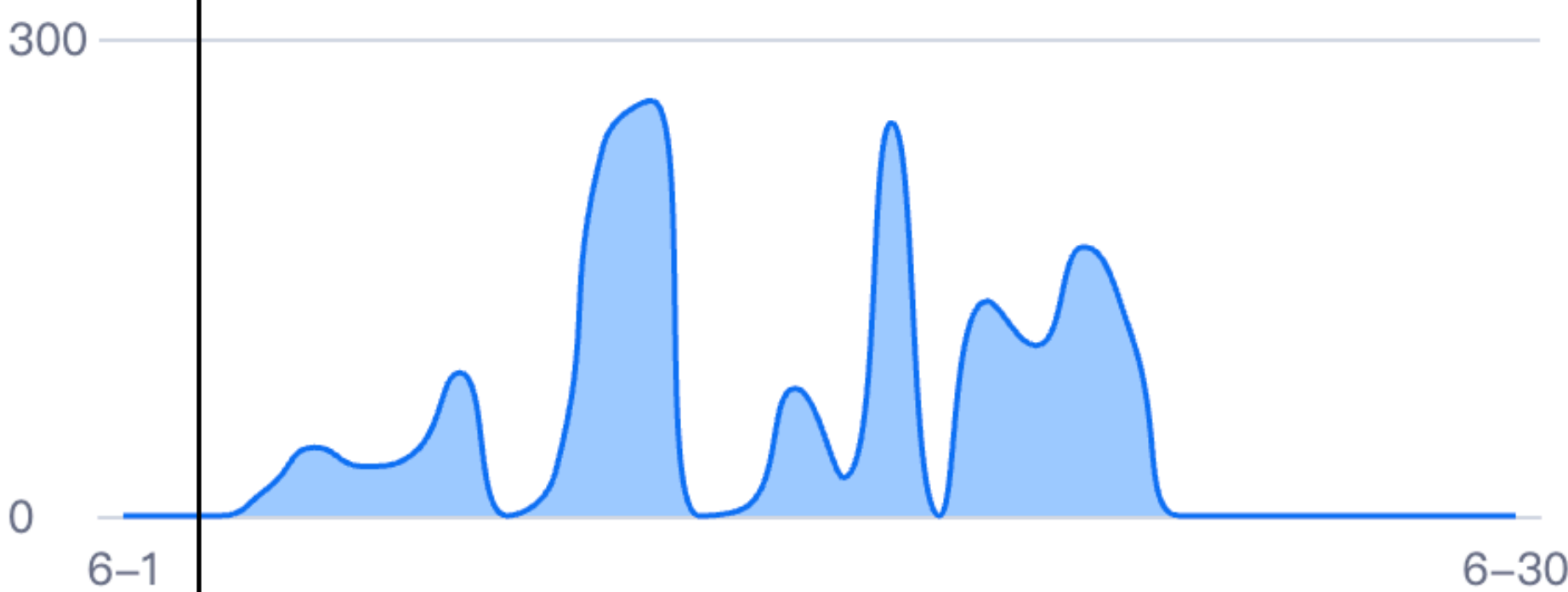
消费金额 ¥3.55

A complete workflow costs between 0.1¥ and 1¥.

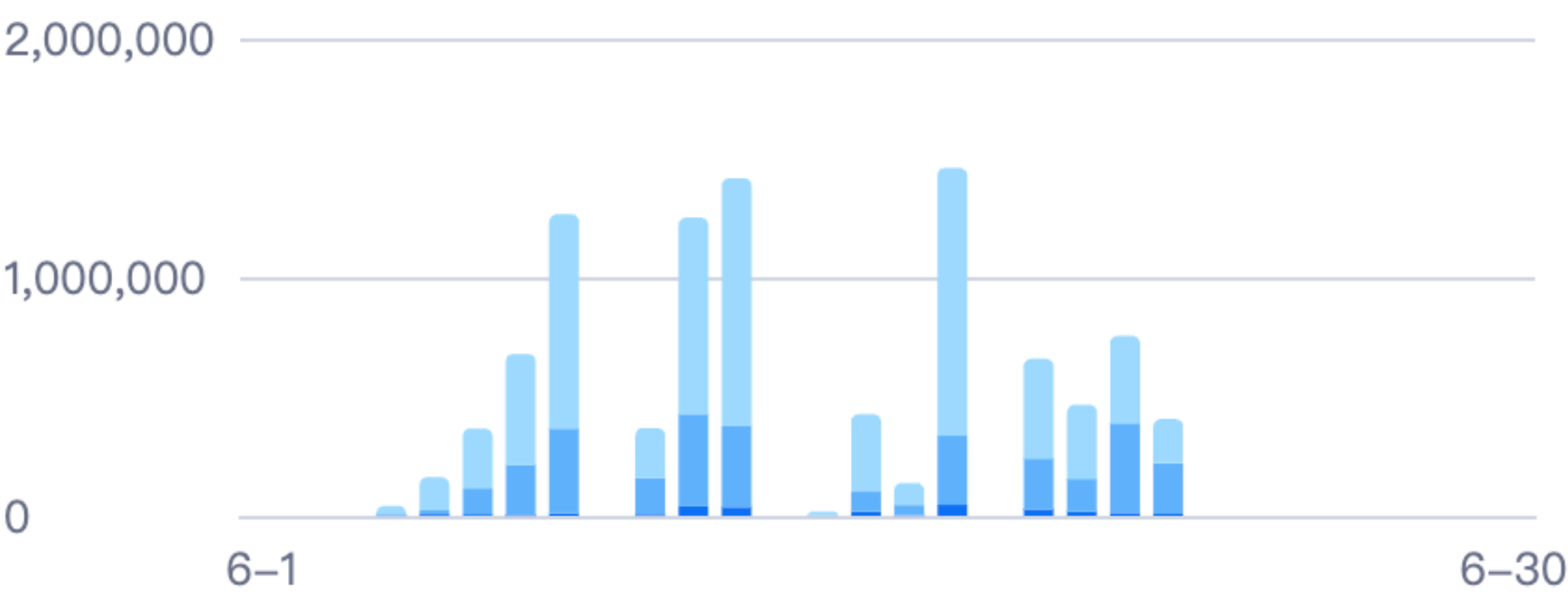


deepseek-v4-flash

API 请求次数 1,621



Tokens 9,861,889



Since calculation code is very fixed, only "dumb" LLM model needs to be used.

LaMET-Agent: installation and usage

<https://github.com/Greyyy-HJC/lamet-agent/tree/main>

Installation

```
git clone https://github.com/Greyyy-HJC/lamet-agent.git
cd ./lamet-agent
deactivate 2>/dev/null || true
/opt/homebrew/bin/python3.13 -m venv .venv_lamet
source .venv_lamet/bin/activate
python --version
python -m pip install -U pip
python -m pip install -e "[dev,analysis]"
```

Find your python(>3.10)

Configuration (~/.zshrc)

```
source ?/lamet-agent/.venv_lamet/bin/activate
export DEEPSEEK_API_KEY='sk-xxx'
```

Usage

```
lamet-agent validate *.json
lamet-agent plan *.json --model deepseek --backend api
lamet-agent run *.json --model deepseek --backend api --report_language ch -v
```



LQCDMaster v.s. LaMET-Agent

	LQCDMaster	LaMET-Agent
核心定位	自然语言物理任务下的科研代码生成器	LaMET/LQCD 分析的可复现分析脚手架
具体目标	上游关联函数	下游数据分析
难点	LLM 探索算符构造方式，调整 Correlator 收缩并对齐指标	LLM 每次数值实现稳定自治，并产生相对可靠的 Review
语料/价格	1700w lines / 2-10¥	5w lines/ 0.2¥
Agent 架构	两段式 Planner → Executor → Slurm Submit	多 stage DAG runner，每个 stage/job 独立 LLM loop
人工差别	LLM 生成代码自由度高，人工审查多	LLM 只介入拟合选取等步骤，人工审查少

LQCDMaster: installation and usage

https://github.com/sjtu-sai-agents/LQCD_Master#

<https://arxiv.org/pdf/2607.15001>

Installation

```
git clone https://github.com/sjtu-sai-agents/LQCD_Master.git
cd ./LQCD_Master
python -m pip install -r requirements.txt
cat > ./env << 'EOF'
# LLM API Configuration
OPENAI_API_KEY=sk-xxx
OPENAI_BASE_URL=https://api.deepseek.com/v1
OPENAI_MODEL=deepseek-v4-pro
# Web Search Tool (Serper.dev)
SERPER_API_KEY=你的Serper密钥 #联网检索功能
EOF
```

Usage

```
python run.py --task "Calculate proton two-point correlator and three-point correlator with current insertion  $\bar{\psi}\gamma_t\psi$  using sequential source method" --test --non-interactive
```


LQCDMaster: installation and usage

https://github.com/sjtu-sai-agents/LQCD_Master#

<https://arxiv.org/pdf/2607.15001>

python run.py CLI可选参数

```
--task "<text>"/<file>: 直接给自然语言任务  
--test: 只生成产物, 不提交 Slurm  
--run-dir <path>: 手动指定输出目录  
--non-interactive: 自动通过 Planner/Executor checkpoint
```

环境配置信息

```
configs/config.yaml: 默认的集群, 组态, module, QUDA 环境配置  
configs/ensemble_presets.yaml: ensemble 知识库 (看起来默认在昆山上)  
configs/skills.yaml: skills 知识库
```

输出文件

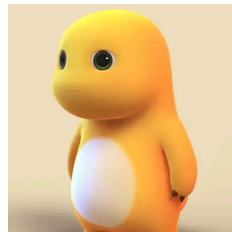
```
runs/<timestamp>/executor_v1/main.py: 主程序  
runs/<timestamp>/executor_v1/submit_test.sh: 测试提交脚本  
runs/<timestamp>/executor_v1/submit_full.sh: 正式提交脚本  
runs/<timestamp>/planner: 最终通过的计算计划
```

LQCD Final Solution in 2026

<https://github.com/Greyyy-HJC/lamet-agent/tree/main>
https://github.com/sjtu-sai-agents/LQCD_Master#

2.0¥

DSV4Pro, GPT5.6Sol



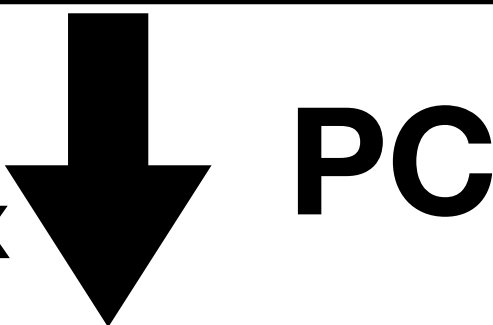
PyQUDA *.py script generated by **LQCDMaster**



∞¥

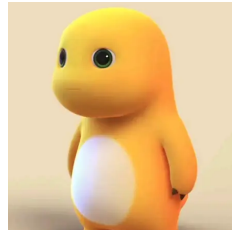


Two-point and Three-point correlators

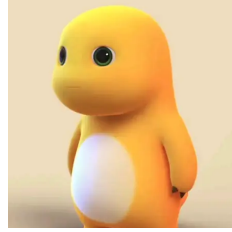


0.2¥ lamet-agent plan or codex

DSV4Flash, GLM-8-fp5

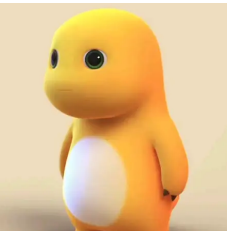


.h5 correlators + *.json can analysis by **LaMET-Agent**



review.md generated by **LaMET-Agent**

0.0¥



Check the results of each step by **Human**

祝论文 发得快

